PROJECT BAYMAX

Baymax

Sheet Samples collection based on Brazil Dam

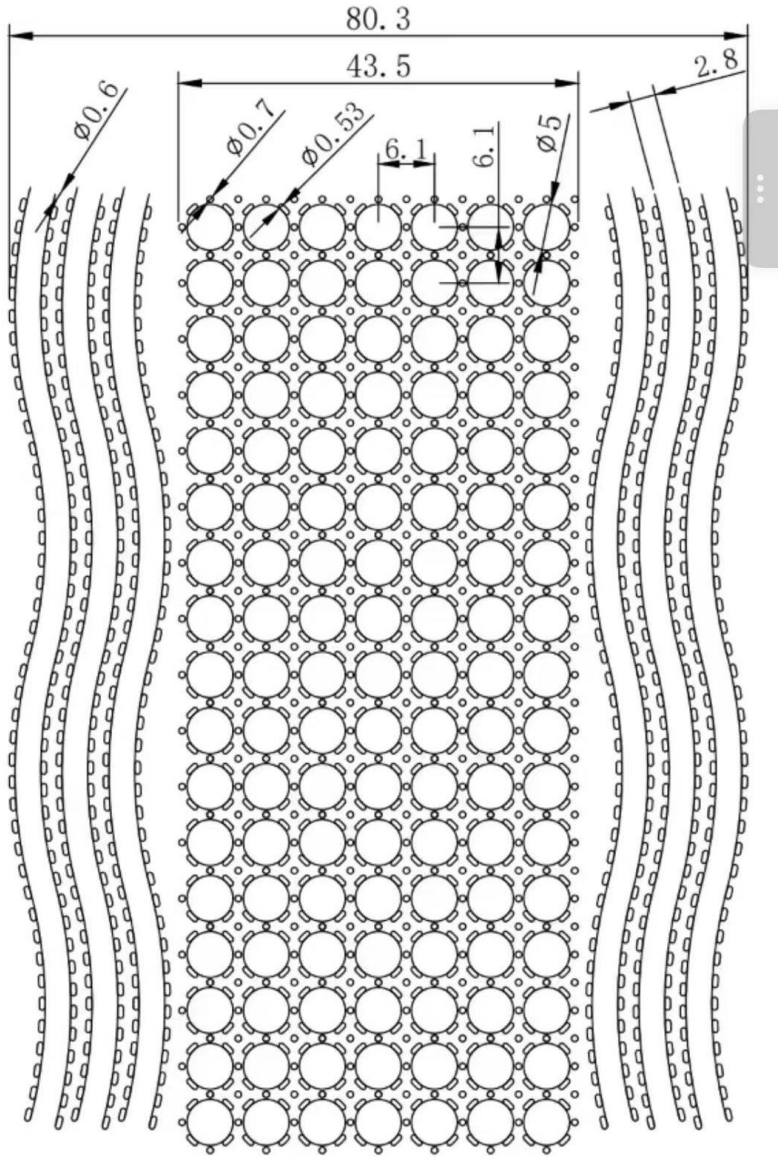
Liner pattern

Oct 14. 2026

Tingting Du



Brazil Dam Liner Pattern

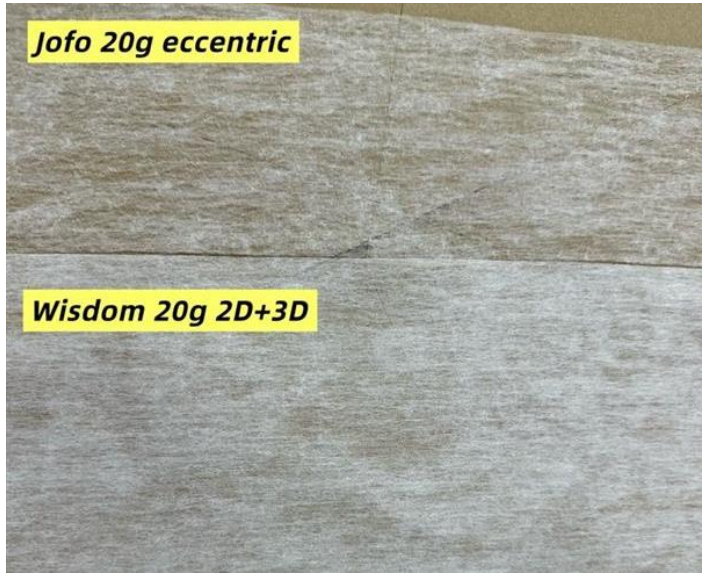


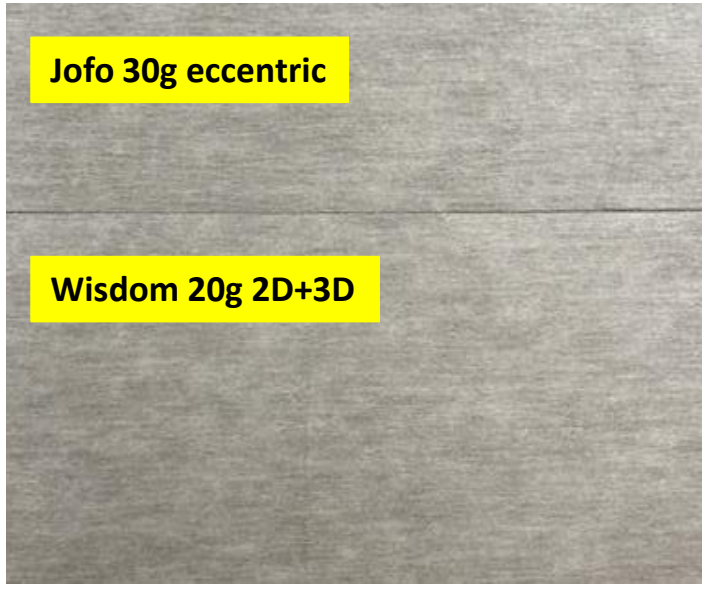
Code List design

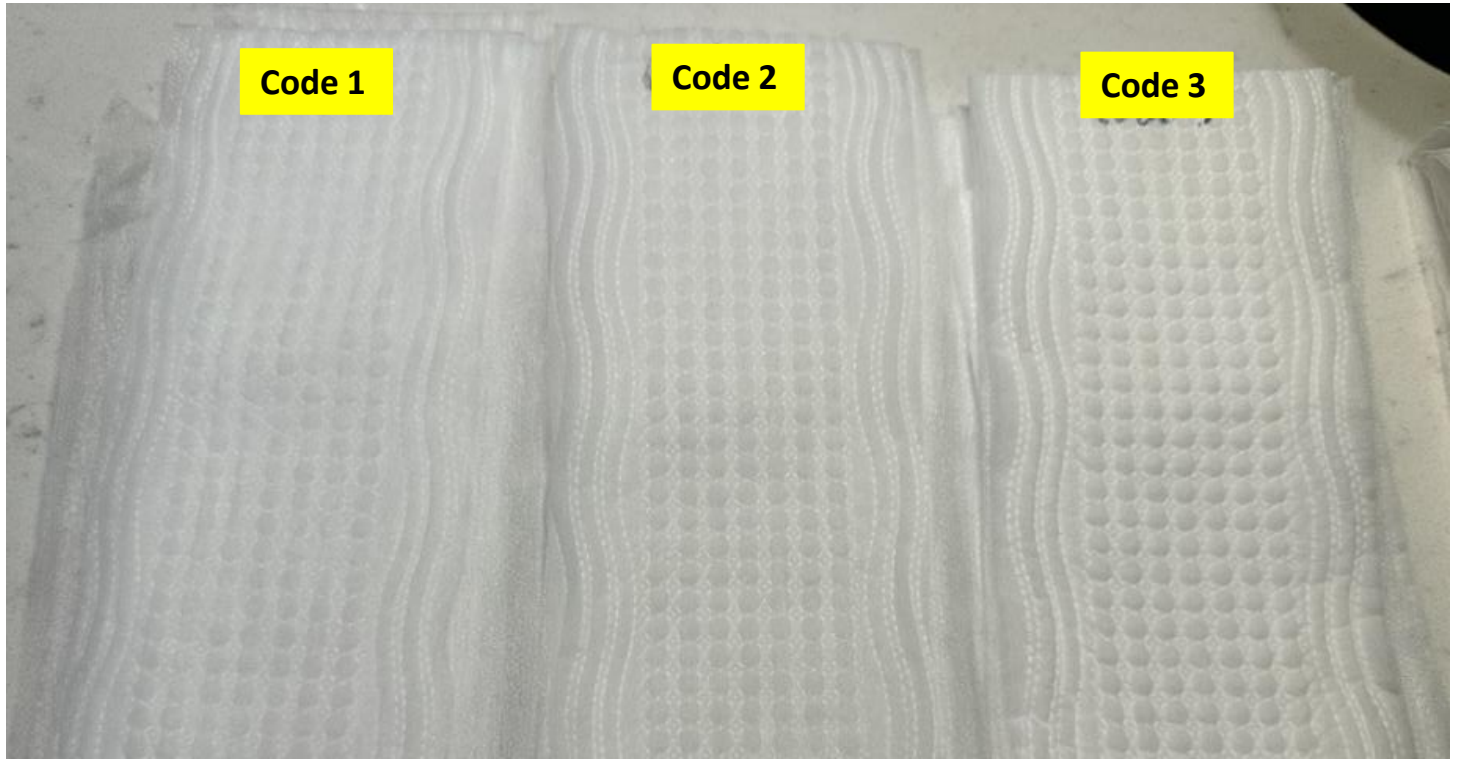
Baymax - New material combination screening			
Code	Exploration Objective	MATERIAL COMBINATION	
		Top Layer (82mm x 259mm)	Bottom Layer (80mm x 259mm)
1	Explore impact of different basis weight on **visual perception and 3D appareance** w/ eccentric fibers	**20 gsm** Upper: 50% 2D，Semi philic Lower: 50% 6D eccentric, Full philic	**10gsm,** Full philic SB BICO (PE/PP) or PP or low melting PP
2		**30 gsm** Upper: 50% 2D,Semi philic Lower: 50% 6D eccentric, Full philic	Based on the result of code 1
3		**40 gsm** Upper: 50% 2D，Semi philic Lower: 50% 6D eccentric, Full philic	Based on the result of code 1
4	Explore impact of different fibers types in dual top layer on **visual perception and 3D appareance**, maintaining 2D on Upper layer for softness and chaging from 3D to 8D w/o eccentric fiber on Lower layer for 3D effect.	30 gsm Upper:50% 2D，Semi philic Lower: 50% **3D**, Full philic	Based on the result of code 1
5		30 gsm Upper: 50% 2D，Semi philic Lower: 50% **6D**, Full philic	Based on the result of code 1
6	Explore TABCW in Bottom Layer, comparing to SB	**Depend on the 3D effect from previous codes but Except the 40gsm one**	TABCW 4D 18 gsm, Full philic
CN & FEI Code			
7	If higher ratio of Lower layer can improve 3D effect	30 gsm Upper: 10gsm 2D，Semi philic Lower: 20gsm **6D**, Full philic	Based on the result of code 1

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- The 20gsm with eccentric fiber shows uneven fiber distribution, which may negatively affect the 3D effect.
- The 30gsm with eccentric fiber shows same fiber distribution with normal 20gsm TABCW



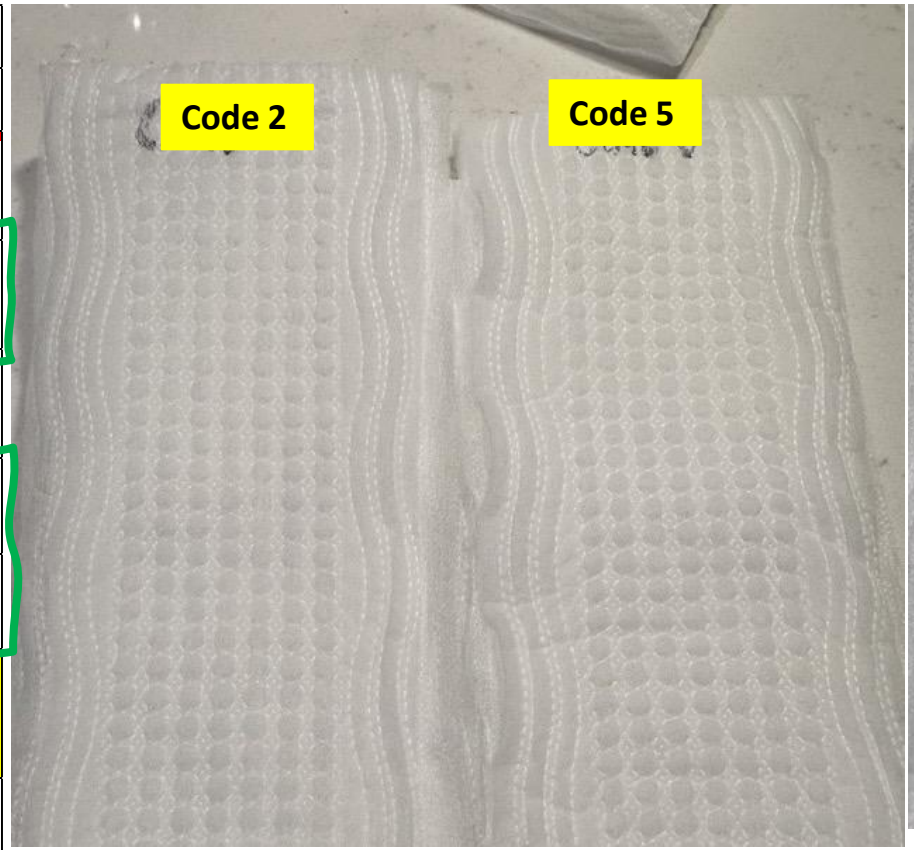


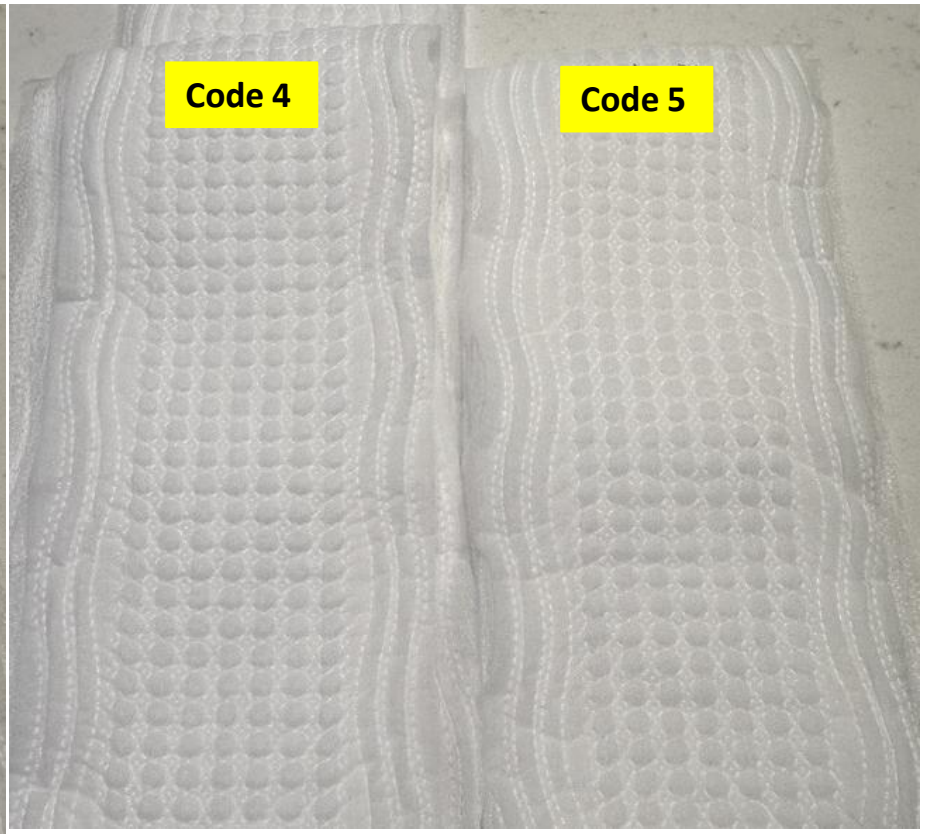


- The Code 3 (40gsm with eccentric fiber) shows the best 3D effect
- For the bottom layer, the lamination strength ranks as Bico > Low melting PP> Normal PP. However the lamination strength between TABCW and normal PP is also acceptable and there is no significant difference in the 3D effect among the three bottom-layer options.
- Due to the necking of top-layer it cannot be guaranteed that 3 Dam on each side, only 2 Dam on each side. Therefore, the top layer width should be 90 mm.

Brazil Dam Liner

Code	MATERIAL COMBINATION	
	Top Layer (82mm x 259mm)	Bottom Layer (80mm x 259mm)
1	**20 gsm** Upper: 50% 2D，Semi philic Lower: 50% 6D eccentric, Full philic	**10gsm,** Full philic Normal PP
2	**30 gsm** Upper: 50% 2D,Semi philic Lower: 50% 6D eccentric, Full philic	10gsm, Full philic Normal PP
3	**40 gsm** Upper: 50% 2D，Semi philic Lower: 50% 6D eccentric, Full philic	10gsm, Full philic Normal PP
4	30 gsm Upper:50% 2D，Semi philic Lower: 50% **3D**, Full philic	10gsm, Full philic Normal PP
5	30 gsm Upper: 50% 2D，Semi philic Lower: 50% **6D**, Full philic	10gsm, Full philic Normal PP
6	**40 gsm Upper: 50% 2D，Semi philic Lower: 50% 6D eccentric, Full philic**	TABCW 4D 18 gsm, Full philic
7	30 gsm Upper: 10gsm 2D，Semi philic Lower: 20gsm **6D**, Full philic	10gsm, Full philic Normal PP
8	**40 gsm Upper: 50% 2D，Semi philic Lower: 50% 6D eccentric, Full philic**	10gsm, Full philic Bico



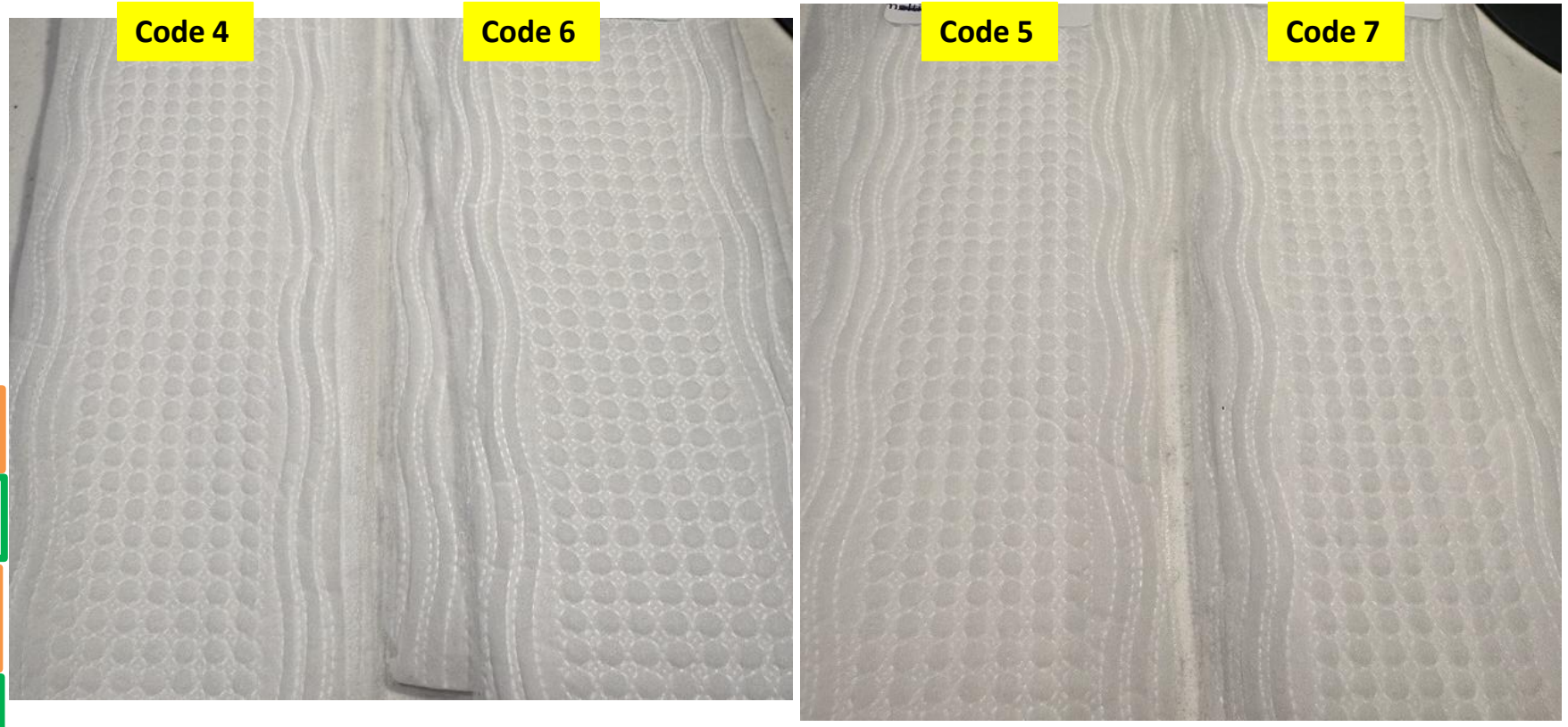


- Code 2 vs Code 5: no significant different in 3D effect
- Code 4 vs Code 5: no significant different in 3D effect

- The effect of eccentric fibers on the 3D effect was not significant in this trial.
- Increasing the lower-layer fiber denier from 3D to 6D did not significantly affect the 3D effect in this trial.

Brazil Dam Liner

Code	MATERIAL COMBINATION	
	Top Layer (82mm x 259mm)	Bottom Layer (80mm x 259mm)
1	**20 gsm** Upper: 50% 2D，Semi philic Lower: 50% 6D eccentric, Full philic	**10gsm,** Full philic Normal PP
2	**30 gsm** Upper: 50% 2D,Semi philic Lower: 50% 6D eccentric, Full philic	10gsm, Full philic Normal PP
3	**40 gsm** Upper: 50% 2D，Semi philic Lower: 50% 6D eccentric, Full philic	10gsm, Full philic Normal PP
4	30 gsm Upper:50% 2D，Semi philic Lower: 50% **3D**，Full philic	10gsm, Full philic Normal PP
5	30 gsm Upper: 50% 2D，Semi philic Lower: 50% **6D**, Full philic	10gsm, Full philic Normal PP
6	**40 gsm Upper: 50% 2D，Semi philic Lower: 50% 6D eccentric, Full philic**	TABCW 4D 18 gsm, Full philic
7	30 gsm Upper: 10gsm 2D，Semi philic Lower: 20gsm **6D**, Full philic	10gsm, Full philic Normal PP
8	**40 gsm Upper: 50% 2D，Semi philic Lower: 50% 6D eccentric, Full philic**	10gsm, Full philic Bico



- Code 4 vs Code 6: no significant different in 3D effect
- Code 5 vs Code 7: no significant different in 3D effect

- The bottom-layer material had no significant effect on the 3D effect.
- Increasing the proportion of higher-denier fibers in the dual-carded lower layer did not significantly affect the 3D effect in this trial.



Summary

Code	MATERIAL COMBINATION		Comparative	Summary
	Top Layer (82mm x 259mm)	Bottom Layer (80mm x 259mm)		
1	**20 gsm** Upper: 50% 2D，Semi philic Lower: 50% 6D eccentric, Full philic	**10gsm,** Full philic Normal PP	1，2，3	**Not recommended, as the uneven fiber distribution in the material results in a very poor 3D effect.**
2	**30 gsm** Upper: 50% 2D,Semi philic Lower: 50% 6D eccentric, Full philic	10gsm, Full philic Normal PP	1，2，3	The 3D effect is acceptable
3	**40 gsm** Upper: 50% 2D，Semi philic Lower: 50% 6D eccentric, Full philic	10gsm, Full philic Normal PP	1，2，3	Delivers the best 3D effect among the three codes, but has no cost advantage.
4	30 gsm Upper:50% 2D，Semi philic Lower: 50% **3D**, Full philic	10gsm, Full philic Normal PP	4，5	- The 3D effect is acceptable - The lower layer with higher denier fibers did not show improvement for 3D effect. - The lower layer with eccentic fibers did not show improvement for 3D effect.
5	30 gsm Upper: 50% 2D，Semi philic Lower: 50% **6D**, Full philic	10gsm, Full philic Normal PP	2，4，5	
6	**40 gsm** **Upper: 50% 2D，Semi philic** **Lower: 50%3D eccentric, Full philic**	TABCW 4D 18 gsm, Full philic	4，6	- The choice of material for the bottom layer has little impact on the 3D effect of the top layer
7	30 gsm Upper: 10gsm 2D，Semi philic Lower: 20gsm **6D**, Full philic	10gsm, Full philic Normal PP	5，7	- The 3D effect is acceptable - More higer denier fiber in lower layer did not show improvement for 3D effect.
8	**40 gsm** **Upper: 50% 2D，Semi philic** **Lower: 50% 3D eccentric, Full philic**	10gsm, Full philic Bico	4，8	- If the lamination strength of normal PP with TABCW cant meet Brazil team's requirements, Bico Spunbond would be a good backup option.

- The lamination strength ranks as Bico > Low melting PP> Normal PP. However the lamination strength between TABCW and normal PP is also acceptable
- The top layer width should be 90 mm to make sure 3 Dams in each side